

A Monolithic Self-Sensing Precision Stage: Design, Modeling, Calibration, and Hysteresis Compensation

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Abstract—This paper presents the design, modeling, calibration, and hysteresis compensation of a self-sensing precision stage used for active vibration isolation. The stage prototype is monolithically fabricated from the spring steel by wire electrical discharge machining process. Aiming to achieve an extremely compact structure, we design and fabricate a self-sensing actuator called smart piezo stack which is capable of not only generating high-resolution displacement but also monitoring the dynamic characteristics of the proposed stage. By means of the finite-element analysis and experimental measurements, we reveal and quantitatively analyze the crosstalk phenomenon in the smart piezo stack. After calibrating the sensitivity of the smart piezo stack experimentally, the dynamics model of the proposed stage is established. Furthermore, a nonlinear autoregressive moving average with exogenous inputs model based on backpropagation neural network is proposed to design a nonlinear controller based on adaptive inverse control. The intelligence of the developed controller allows the hysteresis in the stage which results from the embedded smart piezo stack to be directly compensated for without dynamics modeling in advance. Experimental validation of the adaptive inverse controller is conducted and the results demonstrate the effectiveness of the proposed mechanism and the developed control system.

Index Terms—Calibration, hysteresis compensation, micromotion stage, nonlinear control, piezoelectric actuator, piezoelectric sensor.

I. INTRODUCTION

RECENTLY, precision active vibration isolation is becoming more and more important in various fields of research. Most vibration-sensitive equipment and precision inspection facilities cannot work without vibration isolation; such instruments include atomic force microscope (AFM), scanning probe microscope (SPM), lithography tool, ultraprecision machine tool, satellite and so on. These devices are very susceptible to the disturbances transmitted principally from ground vibration, acoustic noise, or direct mechanical disturbance which result in a degradation in performance [1], [2].

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Strategies to improve the dynamic performance of precision equipment belong to the field of position control and vibration control. The aim of position control is prescribing certain desired motions, whereas the aim of vibration control is suppressing unwanted motions. As an active method to control dynamics of the system, position control and vibration control rely on the use of sensors, actuators and control electronics [3]. In order to achieve high accuracy in position control and vibration control, displacement sensors with high resolution to monitor the motion of the stage are adopted, such as capacitive sensors, inductive sensors, laser interferometers, and optical reflective sensors [4]. However, these sensors are typically expensive and physically large, making them difficult to be installed into a fairly compact structure [5], [6]. In many cases, the use of lightweight and flexible structures makes a system vulnerable to vibration which may cause a system malfunction [7]. The motivation of our research is to design a monolithic stage applied to vibration-sensitive equipment or component for active vibration isolation without additional commercial sensors.

Flexure-based and piezoelectric stack-actuated compliant micromotion stage [8]–[10] is a promising way to accomplish this objective. Recently, some compliant micromotion stages have been proposed and successfully implemented to achieve precision motion control. Particularly, Li and Xu [11] presented a parallel-kinematic XY stage for micro/nanomanipulation. Based on the identified plant transfer function of the system, the H_∞ robust control combined with the repetitive control was adopted to compensate for the unmodeled piezoelectric nonlinearity. Kenton and Leang [12] designed and controlled a serial-kinematic nanopositioning stage that offered approximately $9\ \mu\text{m} \times 9\ \mu\text{m} \times 1\ \mu\text{m}$ range of motion and kilohertz bandwidth. The prototype stage was integrated with a commercial SPM to demonstrate its performance. Yong *et al.* [13] designed a piezoelectric stack-actuated XY nanopositioning stage which combined the ability to scan over a relatively large range with high scanning speed. Nonlinearities such as hysteresis due to the use of piezoelectric stacks for actuation were minimized using charge actuation. In addition, recent development of the smart structure that integrates sensor with actuator [14]–[16] can allow the micromotion stage to be more compact and low cost.

A major problem that compromises the precision of these piezo-actuated systems arises from the nonlinearity mainly due to the hysteresis. In order to fulfill the requirement of precision active vibration isolation, hysteresis ought to be compensated for by an appropriate control scheme [4]. The compensation for hysteresis in micro/nanomotion stage is usually achieved by charge control or model-based voltage control. The main

benefit achieved using charge control is the reduction of hysteresis and drift, whereas the disadvantages include the need for additional electric circuits and the increased complexity of the control hardware [17]. Thus, voltage control is widely used for controlling micro/nanomotion stage including feedforward voltage control, where accurate nonlinear models are typically used and feedback voltage control, where various sensors are employed [18]–[21].

In this paper, a new monolithic precision stage for active vibration isolation is presented. In order to achieve a compact structure, a smart piezo stack is embedded in the proposed stage so as to avoid the usage of additional commercial sensors. Through finite-element analysis (FEA) and experimental measurements, a crosstalk phenomenon in the smart piezo stack is revealed and quantitatively analyzed. Then, we calibrate the sensor in the smart piezo stack by means of experiments and construct the dynamics model of the proposed stage. Additionally, in order to compensate for the hysteresis in the proposed stage, we design a nonlinear controller based on adaptive inverse control. Compared with most existing controllers for hysteresis compensation, a distinctive feature of the developed controller is that it allows the hysteresis due to the smart piezo stack to be directly compensated for without modeling. Thus, the process of parameters identification can be omitted. Experiments are conducted to evaluate the performance of the developed control system.

The remainder of this paper is organized as follows. In Section II, an architecture description of the proposed stage is addressed. In Section III, we elaborate the design, modeling, and calibration of the proposed stage, and the crosstalk phenomenon which occurs in the smart piezo stack is analyzed by means of FEA and experiments. In Section IV, an adaptive inverse controller is designed to compensate for the hysteresis in the proposed stage. In Section V, the experimental implementation is carried out to validate the performance of the developed controller. Finally, conclusions are given and future works are discussed in Section VI.

II. ARCHITECTURE DESCRIPTION

In this paper, a flexure-based compliant precision stage is designed for active vibration isolation. The compliant mechanism delivers motions by making use of the elastic deformations of notch hinges instead of conventional mechanical joints, which renders the stage with merits of free of backlash, zero friction, repeatable motion, and vacuum compatibility. Besides, piezoelectric stack actuators are commonly used to drive flexure-based stages due to their capability of positioning with nanometer level resolution, large blocking force, high stiffness, and rapid response characteristics [22], [23].

A. Mechanism Design

The compliant mechanism design plays a key role in the development of the monolithic precision stage. Some recent researches on the design of flexure-based nanopositioning system are presented in [24]–[26]. Fig. 1 shows the prototype of the proposed monolithic self-sensing precision stage for active

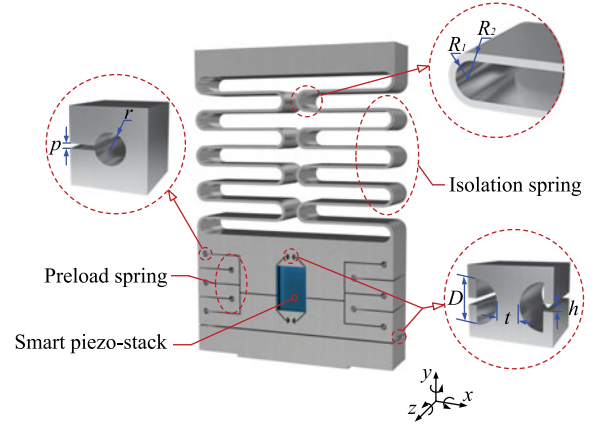


Fig. 1. Monolithic self-sensing precision stage for active vibration isolation.

vibration isolation. The body of the stage is manufactured using the electrical discharge machining (EDM) process to create a monolithic structure. It can be divided into two main parts: 1) the flexure-based isolation spring which is used for passive vibration isolation and 2) the active part which is used for precision position control and vibration control in an active way. In order to allow the proposed stage to feature a compact structure, we design a smart piezo stack which, on the one hand, could drive the proposed stage with high precision, on the other hand, is capable of monitoring the dynamic characteristics of the proposed stage. It can be seen from Fig. 1 that the active part is in series with the isolation spring. Unlike actuators with low stiffness such as Lorentz motors, the piezoelectric actuators are usually installed in series with the passive spring because of their high stiffness and high actuating force. Schubert *et al.* [27] presented a stiff actuator active vibration isolation system which typically adopted this series structure. In this paper, the smart piezo stack is embedded in the proposed stage as an active element for both sensing and actuating. However, owing to the brittle nature of ceramics material, the smart piezo stack should preferably not be exposed to the pulling force. This implies that for dynamic application in which a certain pulling force capacity is desirable, the smart piezo stack should be subjected to a compressive preload in order to withstand tensile loads. Therefore, we design two accordion springs which consist of a series of flexure hinges to provide preload for the smart piezo stack. Additionally, the smart piezo stack should also avoid being tilted and sheared. In other words, the smart piezo stack should only be loaded axially, i.e., along the y-axis. To ensure this, the compliant flexures are designed to have high stiffness in the actuation direction, while being sufficiently soft in other directions. The dimensions of the active part of the proposed stage are shown in Fig. 2 and tabulated in Table I. All the hinges are designed to have identical dimensions for convenient fabrication. It can be seen that the presence of a couple of 45° and 135° cuts end up in a pair of flexure hinges and a longitudinal flexure hinge is equipped. This decoupling mechanism possesses a high stiffness in its working direction and a proper compliance in the transverse direction. Herein, the material is assigned as 60Si2Mn spring steel with its main parameters described in Table II.

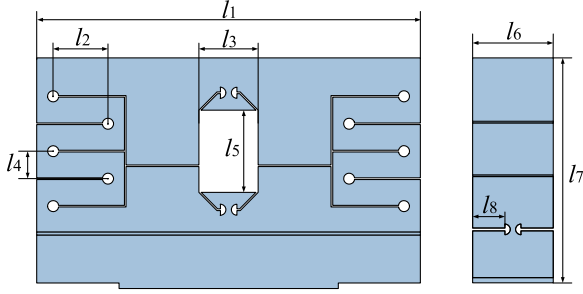


Fig. 2. Dimensions of the active part of the proposed stage.

TABLE I
ARCHITECTURAL PARAMETERS

Parameter	Value(mm)	Parameter	Value(mm)
r	1	l_2	10
p	0.3	l_3	11
R_1	2.5	l_4	5
R_2	3.5	l_5	13
D	2	l_6	15
t	1	l_7	42
h	0.3	l_8	6
l_1	70	-	-

TABLE II
MATERIAL PARAMETERS OF FLEXURE-BASED STAGE

Parameter	Value	Unit
Young's modulus	206	GPa
Yield strength	1176	MPa
Poisson's ratio	0.31	-
Density	7850	Kg/m ³

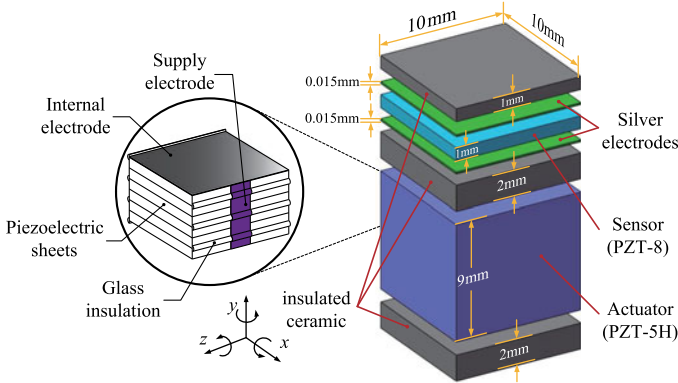


Fig. 3. Schematic of the smart piezo stack.

B. Fabrication of the Smart Piezo Stack

The smart piezo stack is a crucial component of the monolithic self-sensing precision stage as it not only provides active ways to achieve position control and vibration control, but also acts as sensors of the proposed stage. The schematic and the photograph of the smart piezo stack are shown in Fig. 3 and Fig. 4, respectively. The actuator is a multilayer stack which consists of thin piezoelectric sheets. It can be seen that the total height of

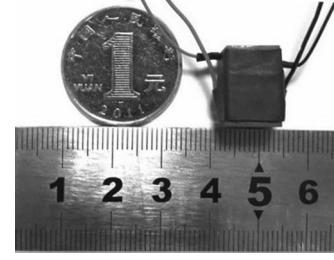


Fig. 4. Photograph of the smart piezo stack.

TABLE III
PARAMETERS OF PIEZOELECTRIC MATERIAL

Material types		PZT-8	PZT-5H
Dielectric constant	E_{33}^T/E_0	1000	3200
Dissipation factor [%]	$\tan \delta$	0.2	1.1
Piezoelectric charge constants [pC/N]	d_{33}	150	540
	d_{31}	-25	-280
Mechanical Q	Q_m	800	60
Elastic constants [$\times 10^{10}$ N/m]	Y_{11}^E	8.6	6
	Y_{33}^E	7.1	4.3

the piezo stack is 15.3 mm, whereas the height of the recess for the smart piezo stack l_5 is 15 mm. Thus, the proposed stage can provide a certain amount of preload when the smart piezo stack is recessed. In this case, all the piezoelectric materials are used in d_{33} -operation where the “three”-direction refers to the direction of polarization. It implies that for the smart piezo stack, we are interested in its abilities of actuating and sensing along the poling axis. The parameters of the piezoelectric materials are tabulated in Table III.

III. MODELING, ANALYSIS, AND CALIBRATION OF THE STAGE

The dynamics of piezoelectric material can be explained by a general electromechanical model, which includes the electrical aspect and the mechanical aspect, as well as their mutual interaction [28], [29]. In order to obtain an accurate physical model, the smart piezo stack needs to be experimentally identified. Thus, we calibrate the sensitivity of the sensor in the smart piezo stack via experiments. It is worth mentioning that because we develop an adaptive inverse control strategy for hysteresis compensation as presented in Section IV, the identification of the actuator could be avoided. In addition, we reveal the crosstalk phenomenon which exists in the smart piezo stack. Herein, we adopt both the FEA and experiments to estimate the influence of this crosstalk phenomenon on the dynamics of the proposed stage. Taking into account the interaction of the smart piezo stack and the flexure-based stage, the dynamics model of the active part of the proposed stage is established.

A. Modeling of the Smart Piezo Stack

The piezoelectric effect is an effect in which energy is converted between mechanical and electrical form. When a pressure is applied to the smart piezo stack, the resultant mechanical deformation will result in an electrical charge. In order to utilize

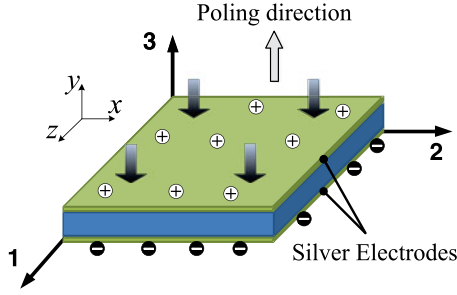


Fig. 5. Schematic of the sensor in smart piezo stack.

this physical principle to make a sensor, we need to measure the surface charge on the piezoelectric sensor layer. As shown in Fig. 5, two silver electrodes sandwich the piezoelectric sensor to form a capacitor. When the smart piezo stack is subjected to a pressure along y-axis, a positive charge accumulates on the top electrode and a negative charge accumulates on the bottom electrode of the sensor. The generated charge is amplified by a charge amplifier whose capacitance C is set to 10^4 pF. Thus, the generated voltage V_s of the piezoelectric force sensor due to the load force F_s can be calculated as [8]

$$V_s = \frac{d_{33} F_s}{C}. \quad (1)$$

It follows from (1) that there exists a linear relationship between the generated voltage and the load force. Thus, the smart piezo stack is capable of acting as transducer that turns force, or mechanical stress into electrical charge which in turn can be converted into a voltage.

Alternatively, when a voltage is applied to the actuator part, the resultant electric field would cause a deformation of the smart piezo stack. The piezoelectric effect is understood as the electromechanical interaction between the mechanical and the electrical state in the piezoelectric material. The converse piezoelectric effect, however, is known to suffer from nonlinearities, due to the so-called “ferroelectric” nature which leads to the rate-dependent hysteresis [30]–[32]. Herein, we define the nonlinear relationship between the applied voltage and the output displacement along y-direction of the smart piezo stack as

$$x_a = f(V_a) \quad (2)$$

where V_a and x_a denote the applied voltage and the output displacement of the smart piezo stack, respectively; $f(\cdot)$ denotes the nonlinearity due to the hysteresis. From (1) and (2), it follows that the smart piezo stack can be considered as an intelligent element which is capable of sensing a pressure applied on it and generating a corresponding displacement.

B. Calibration of the Piezoelectric Force Sensor

Without mapping the fabricated sensor reading into the corrected value, it cannot be used in practical application. Herein, we calibrate the fabricated sensor by determining the accurate relationship between voltage and force experimentally with the setup as shown in Fig. 6. The hardware setup mainly consists

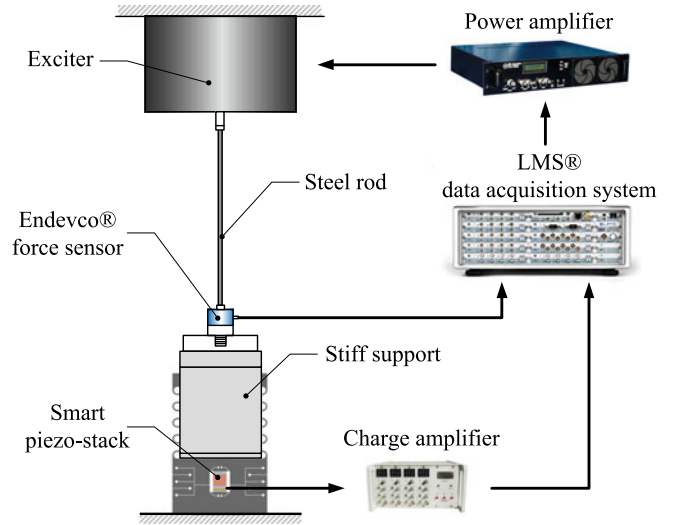


Fig. 6. Connection scheme of the hardware setup for the calibration of the sensor in smart piezo stack.

of an exciter (Modal 110, made by MB Dynamics GmbH), a power amplifier (MB500VI, made by MB Dynamics GmbH), a commercial force sensor (Model 2311-10, made by Endevco Corporation) with sensitivity of 10.01 mV/lbf, a charge amplifier (BZ2105, made by Bdhland Corporation), and an LMS data acquisition system with analog output module (LMS SCADAS III, made by LMS Corporation). The influence of isolation spring is eliminated by a stiff support. The analog output module of LMS data acquisition system sent signal to the exciter via the power amplifier, and the exciter in turn vertically excited the proposed stage through a flexure steel rod. This excitation resulted in a corresponding force that was applied to both commercial force sensor and the fabricated sensor. During the calibration process, the signal from the fabricated sensor was amplified by the charge amplifier and then recorded by the LMS data acquisition system; meanwhile, the measured signal of the commercial force sensor which can be considered as the correct force signal was also recorded by the LMS data acquisition system. As the forces applied to the commercial force sensor and the fabricated piezoelectric sensor were identical, the sensitivity of the fabricated sensor can be calibrated by comparing these two recorded signals.

Fig. 7 shows a photograph of the calibration process of the sensor in the smart piezo stack. In this case, we adopted a chirp signal with frequency increasing from the lower bound of the band to the higher. Utilizing LMS Test.Lab software, we obtained the measured frequency response which describes the relationship between the correct force measured by the commercial force sensor and the output voltage of the charge amplifier as shown in Fig. 8. It can be seen that except for two minor peaks that probably result from mechanical structure, the magnitude of the frequency response function is nearly constant up to 100 Hz. Thus, we can consider the sensitivity of the sensor to be invariable over this frequency range. Due to the capacitive source impedance, the piezoelectric sensor cannot measure signal with very low frequency. According to the measured

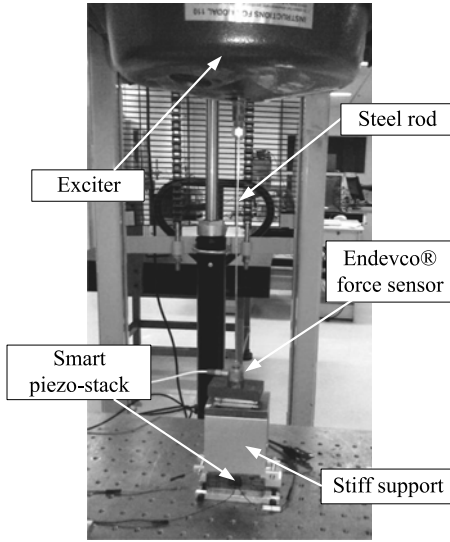


Fig. 7. Photograph of the calibration process of the sensor in the smart piezo stack.

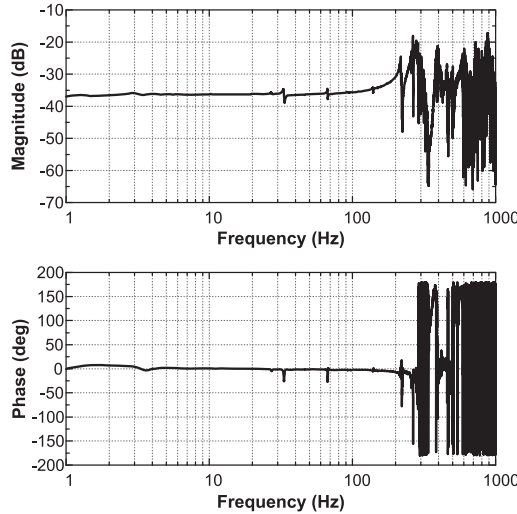


Fig. 8. Measured frequency response of the stage. The input is the force measured by the commercial force sensor and the output is the output voltage of the charge amplifier.

frequency response, the force sensitivity of the smart piezo stack is 0.015381 V/N in a frequency range from 1 to 100 Hz.

C. Crosstalk Between Sensor and Actuator

In the previous modeling and analysis, the piezoelectric sensor and actuator are treated as separated elements without interaction. However, this treatment contradicts the integrated configuration of the smart piezo stack. The crosstalk phenomenon between sensor and actuator in the smart piezo stack is exaggeratedly illustrated in Fig. 9. When the actuator is subjected to a positive electric field, on the one hand, the thickness of the actuator will increase due to the piezoelectric coefficient d_{33} which is positive; on the other hand, the cross section of the actuator will decrease due to the material continuity and the piezoelectric coefficient d_{31} which is negative. As the sensor is integrated with

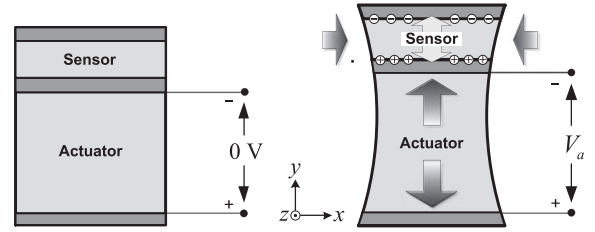


Fig. 9. Illustration of the crosstalk in the smart piezo stack.

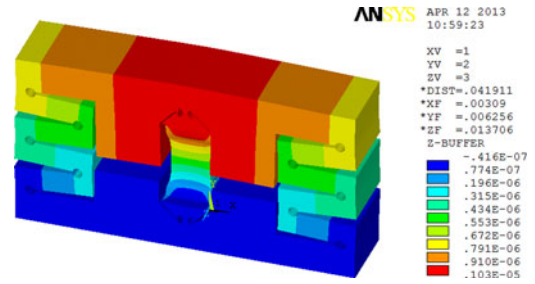


Fig. 10. Deformed FE model of the active part of the stage when 20-V voltage is applied to the smart piezo stack.

the actuator, the decrease of the cross section of the actuator will inherently induce the decrease of the cross section of the sensor, which in turn leads to the increase of the thickness of the sensor also due to the material continuity. From the viewpoint of the piezoelectric sensor, an increase in the thickness corresponds to a tensile force applied along the direction of the thickness, i.e., along the y-axis. Compared with Fig. 5, the polarity of charge generated by the internal crosstalk phenomenon is opposite to the charge generated by the external force along the y-axis. It means that the crosstalk phenomenon leads to a decrease of the amount of charge within the piezoelectric sensor in comparison with the actual value.

Taking advantage of the FEA, we quantitatively estimate the influence of the crosstalk phenomenon in the smart piezo stack. Fig. 10 shows the displacement contour of the stage along y-axis when 20-V voltage is applied to the smart piezo stack. It is obvious that if the smart piezo stack is subjected to an electric field, it also deforms in the lateral directions as a result of its intended elongation. The FEA results shown in Fig. 11 exhibit the influence of crosstalk phenomenon on the sensor. It can be seen that for the isolated smart piezo stack which is not preloaded by the accordion springs, the crosstalk phenomenon gives rise to -0.006636 -V voltage on the sensor when 20-V voltage is applied to the actuator.

Furthermore, we conducted a group of experiments to examine the crosstalk phenomenon and the experimental results are compared with the FEA results. For the proposed stage with embedded smart piezo stack, when sinusoidal voltages with various amplitudes are applied, the amplitudes of the sensor voltage amplified by the charge amplifier are measured as shown in Fig. 12. In addition, for the isolated smart piezo stack, we also applied sinusoidal voltage with various amplitudes and the crosstalk voltages are measured as shown in Fig. 13. As can be seen, the relationship between voltage applied to the isolated smart

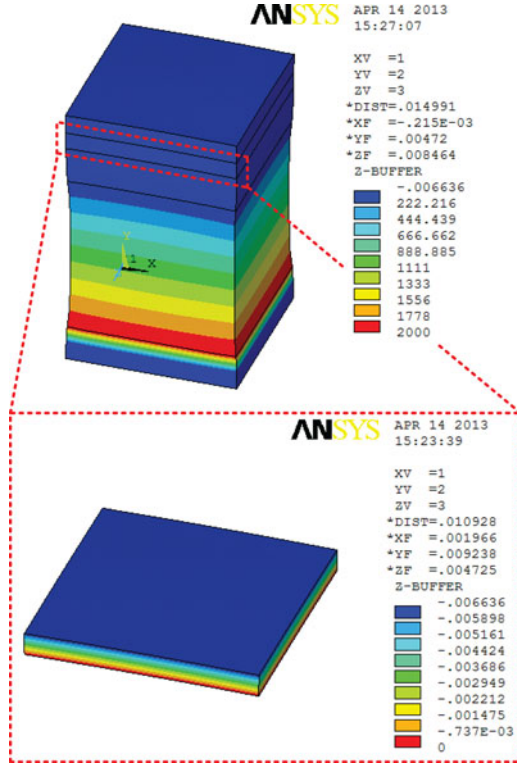


Fig. 11. FEA results of the crosstalk voltage when 20-V voltage is applied to the isolated smart piezo stack.

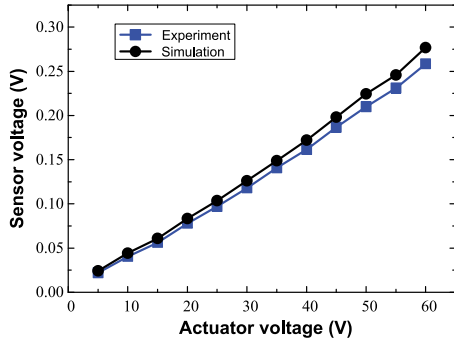


Fig. 12. Relationship between actuator voltage and sensor voltage of an embedded smart piezo stack.

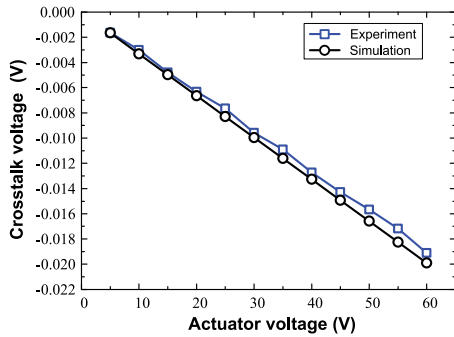


Fig. 13. Relationship between actuator voltage and crosstalk voltage of an isolated smart piezo stack.

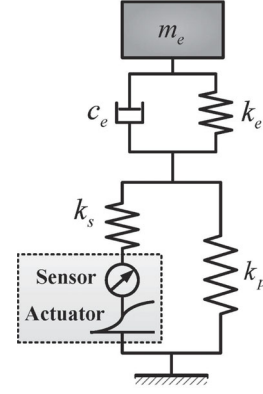


Fig. 14. Graphical model depicting the interaction between smart piezo stack and the active part of the proposed stage.

piezo stack and the crosstalk voltage is linear. In addition, the fact that the crosstalk phenomenon generates negative voltage is in agreement with the analysis as illustrated in Fig. 9. Comparing Fig. 12 with Fig. 13, it is noticeable that the crosstalk voltage accounts for less than 8% of the sensor voltage. This is due to the 2-mm thick insulated ceramic layer which effectively restrains the crosstalk phenomenon in the smart piezo stack. It is worth mentioning that sandwiching a metal ball between the actuator and the sensor could further suppress the crosstalk phenomenon.

D. Dynamics Modeling of the Active Part of Stage

Based on the established model of the smart piezo stack, the system dynamics model of the active part of the proposed stage with embedded smart piezo stack is constructed. From a mechanical viewpoint, the smart piezo stack which comprises a force sensor and a position actuator is subjected to an elastic preload, as shown in Fig. 14. Herein, k_p and k_s denote the stiffnesses of the preload spring and the smart piezo stack, respectively; m_e , c_e , and k_e denote the effective mass, effective damping, and effective stiffness of the active part of the proposed stage, respectively.

Following Fig. 14, the transfer function from the output displacement to the force applied to the smart piezo stack is given by

$$\frac{F_s}{x_a} = \frac{(k_s k_e + k_e k_p + k_p k_s) m_e s^2 + k_p k_s c_e s + k_s k_e k_p}{(k_p + k_s) m_e s^2 + (k_p + k_s) c_e s + k_s k_e + k_e k_p}. \quad (3)$$

According to (1), (2), and (3), the transfer function from the applied voltage to the measured voltage of the smart piezo stack should be

$$\frac{V_s}{V_a} = \left[\frac{(k_s k_e + k_e k_p + k_p k_s) m_e s^2 + k_p k_s c_e s + k_s k_e k_p}{(k_p + k_s) m_e s^2 + (k_p + k_s) c_e s + k_s k_e + k_e k_p} \right] \cdot \frac{d_{33} f(V_a)}{C V_a}. \quad (4)$$

The frequency of the system zeros is

$$\omega_z = \sqrt{\frac{k_s k_e k_p}{m_e (k_s k_e + k_e k_p + k_p k_s)}}. \quad (5)$$

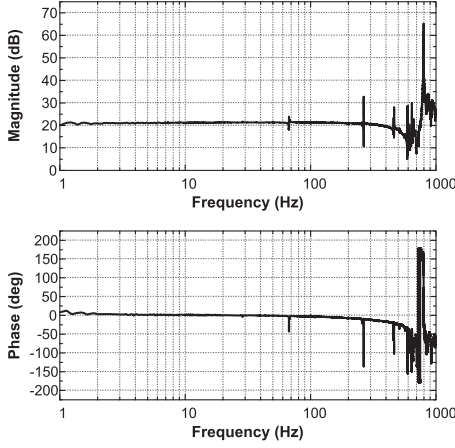


Fig. 15. Measured frequency response of the active part of the stage. The input is output voltage of piezoelectric sensor V_s which is amplified by the charge amplifier and output is the measured displacement x_a .

At the frequency below ω_z , (4) can be simplified to

$$\frac{V_s}{V_a} = \frac{k_s k_p}{k_s + k_p} \cdot \frac{d_{33} f(V_a)}{C V_a}. \quad (6)$$

Herein, we define a crosstalk coefficient p_c as the ratio of crosstalk voltage to actuator voltage. Thus, due to the crosstalk phenomenon, (6) is rewritten as

$$\frac{V_s}{V_a} = \frac{k_s k_p}{k_s + k_p} \cdot \frac{d_{33} f(V_a)}{C V_a} + p_c. \quad (7)$$

For the practical application of the smart piezo stack which is embedded in the proposed stage, when voltage is applied to the actuator, the voltage generated by the sensor comes from two aspects: the force due to the deformation of the preload spring and the crosstalk phenomenon. Thus, it follows from (2) and (7) that

$$V_s = \frac{k_s k_p}{k_s + k_p} \cdot \frac{d_{33} x_a}{C} + p_c f^{-1}(x_a). \quad (8)$$

Following the simulation and experimental results, because the crosstalk voltage only accounts for a small percentage in the total sensor voltage as illustrated previously, we omit crosstalk term and (8) can be rewritten as

$$x_a = \frac{(k_p + k_s)C}{k_p k_s d_{33}} \cdot V_s. \quad (9)$$

According to (9), the displacement of the proposed stage is proportional to the sensor voltage which is amplified by charge amplifier. It means that the displacement of the proposed stage can be measured by the smart piezo stack itself instead of the additional commercial displacement sensor. That is to say, the smart piezo stack can serve as not only a force sensor, but also a displacement sensor. Utilizing LMS data acquisition system, we measure the displacement of the active part of the proposed stage and the sensor voltage which is amplified by charge amplifier. The measured frequency response is shown in Fig. 15. Herein, the displacement of the stage is measured by a laser displacement sensor with 10 nm repeatability (LK-G, made by Keyence Corporation). Similarly, a chirp signal with

increasing frequency is applied to the actuator. According to the experimental results, sensitivity of the smart piezo stack for displacement measuring is $11.6126 \mu\text{m/V}$ in a frequency range from 1 to 100 Hz.

IV. ADAPTIVE INVERSE CONTROLLER DESIGN

Hysteresis inherent in the smart piezo stack deteriorates its positioning accuracy. In addition, nonlinearity is notorious in vibration control problem. The hysteresis is such a kind of complex nonlinearity which may make the vibration control to become more complicated. In this section, a nonlinear autoregressive moving average with exogenous inputs (NARMAX) model based on backpropagation (BP) neural network is proposed to develop an adaptive inverse controller which compensates for the hysteresis in the proposed stage. Particularly, the developed controller allows hysteresis to be directly controlled without dynamics modeling.

A. NARMAX Model Based on BP Neural Network

The NARMAX model is a black-box nonlinear model which provides a powerful representation for time series analysis, modeling, and prediction due to its capability of accommodating the dynamic, complex, and nonlinear nature of time series prediction problems [33]. Due to the powerful capability of nonlinear modeling, the NARMAX model is suitable for hysteresis modeling and compensation of the proposed stage. A special case of the general NARMAX model is represented as [34]

$$y(t) = F[y(t-1), \dots, y(t-n_y), x(t-1), \dots, x(t-n_x)] + E(t) \quad (10)$$

where t is the time variable; $y(t)$, $x(t)$, and $E(t)$ denote the output signal, the input signal, and the system noise, respectively; n_y and n_x denote the associate maximum lags; and $F(\cdot)$ denotes the unknown nonlinear function. Assuming that $E(t)$ has zero mean and a finite variance, an optimal predictor for NARMAX model in (10) is approximately given by

$$y(t) = F[y(t-1), \dots, y(t-n_y), x(t-1), \dots, x(t-n_x)]. \quad (11)$$

Consider both the accuracy and the efficiency, we exploit this optimal predictor to describe the hysteresis. In this case, the unknown nonlinear function $F(\cdot)$ in (11) is modeled by the BP neural network. The starting point of BP neural network is imitating how human brain works. It has a layered structure and each layer consists of a series of adaptive linear neurons (ADALINES). The BP network is basically a gradient decent algorithm designed to minimize the error function in the weights space. During training of the neural network, weights are adjusted to decrease the total error [35].

It follows from (11) that the NARMAX model has self-feedback property which means that the input of neural network contains previous outputs. In fact, the real intelligence of the neural network exists in the weights in ADALINES. Herein, we employ the BP learning algorithm to adjust these weights. The BP learning algorithm takes iterative steps: first, input signal is applied to the network which produces output signal based on

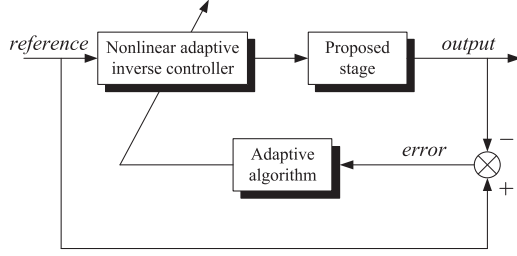


Fig. 16. Block diagram of the adaptive inverse control scheme for the proposed stage.

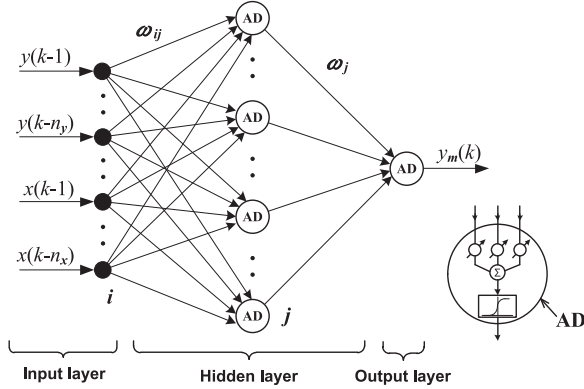


Fig. 17. Structure of the nonlinear adaptive inverse controller.

the current state of ADALINEs' weights; then, this output signal is compared with the known output signal, and a mean-squared error signal is calculated; next, the error value is propagated backward through the neural network and changes are made to the weights for reducing the error signal in each layer. Although BP learning algorithm can be applied to the neural network with any number of layers, experiments show us that one hidden layer can reach sufficient precision and own preferable efficiency.

B. Structure of the Control System

This paper considers the adaptive inverse control, which is based on the concept of dynamic inversion, but an inverse needs not exist. In Fig. 16, the block diagram of the adaptive inverse control scheme for the proposed stage is shown. It can be seen that the control of the proposed stage is achieved by preceding it with a nonlinear adaptive controller whose dynamics are the inverse of the proposed stage. A particular benefit of using this scheme is that the dynamics of the closed-loop system are equal to the dynamics of the open-loop system if the plant model is identical to the plant [36]. The effect of closed-loop response can be obtained in an open-loop feedforward control system by using the feedback inherent in adaptive filtering to find the adaptive inverse controller [37].

For the nonlinear adaptive inverse controller which aims to compensate for the hysteresis in the smart piezo stack, the input is the desired displacement of proposed stage and the output is the applied voltage. In Fig. 17, the structure of the nonlinear adaptive inverse controller is shown. By comparing the measured displacement with the desired displacement, the error is fed back and the proposed adaptive algorithm is used for ad-

justing the controller. Although the hysteresis in the smart piezo stack has no theoretical inverse, the developed controller can still be viewed as an inverse in application, so as to allow the hysteresis to be canceled by its inverse. As shown in Fig. 16, we directly use this nonlinear adaptive inverse controller to control the proposed stage, and the dynamics modeling of hysteresis in the smart piezo stack is avoided. For the nonlinear adaptive inverse controller, we define the input vector as

$$X(k) = \{\bar{y}(k-1), \dots, \bar{y}(k-n_{\bar{y}}); \bar{y}_d(k-1), \dots, \bar{y}_d(k-n_{\bar{y}_d})\} \quad (12)$$

where k denotes the k th sampling time; $\bar{y}(k)$ and $\bar{y}_d(k)$ denote the measured and the desired output displacement of the proposed stage, respectively; $n_{\bar{y}}$ and $n_{\bar{y}_d}$ denote the associate maximum lags. Herein, the limiting function in ADALINEs of the hidden layer and the output layer is log-sigmoid function, that is

$$g(z) = \frac{1}{1 + e^{-z}}. \quad (13)$$

Define the outputs of the input layer as

$$O_i(k) = X_i(k), \quad i = 1, 2, \dots, n_{\bar{y}} + n_{\bar{y}_d}. \quad (14)$$

From Fig. 17, the output of the hidden layer can be found as

$$O_j(k) = g \left[\sum_{i=1}^{n_{\bar{y}} + n_{\bar{y}_d}} \omega_{ij}(k-1) O_i(k) \right], \quad j = 1, 2, \dots, P \quad (15)$$

where P and ω_{ij} denote the number and the weights of ADALINEs in the hidden layer, respectively. According to (13) and (15), the output of the output layer is given by

$$u(k) = \sum_{j=1}^P \omega_j(k-1) O_j(k) \quad (16)$$

where ω_j denotes the weights in ADALINEs in the output layer. In the practical application, $u(k)$ needs to be amplified by a piezoelectric actuator voltage amplifier before applying to the smart piezo stack. After obtaining the measured displacement of the proposed stage, the error between the reference displacement and the measured displacement is given by

$$e(k) = \bar{y}_d(k) - \bar{y}(k). \quad (17)$$

The weights in nonlinear adaptive inverse controller are updated as

$$\begin{aligned} \omega_j(k) &= \omega_j(k-1) + \eta e(k) O_j(k) \\ &\quad + \alpha [\omega_j(k-1) - \omega_j(k-2)] \end{aligned} \quad (18)$$

$$\begin{aligned} \omega_{ij}(k) &= \omega_{ij}(k-1) + \eta e(k) \omega_j(k-1) \\ &\quad \cdot g' \left[\sum_{i=1}^{n_{\bar{y}} + n_{\bar{y}_d}} \omega_{ij}(k-1) O_i(k) \right] O_i(k) \\ &\quad + \alpha [\omega_{ij}(k-1) - \omega_{ij}(k-2)] \end{aligned} \quad (19)$$

where η denotes the study rate; α denotes the dynamic term. Fig. 18 shows the flowchart of the control algorithm. By means

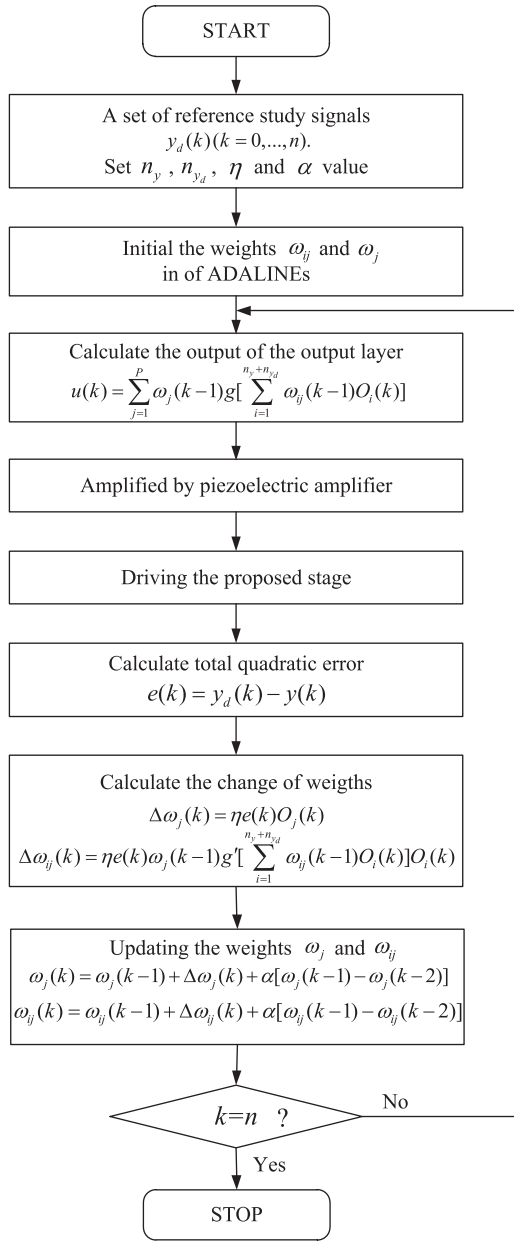


Fig. 18. Flowchart of the control algorithm.

of the strategy to adaptively updating the weights, we can adjust the controller in an online way to compensate for the hysteresis in the proposed stage.

V. CONTROLLER EXPERIMENTAL RESULTS AND DISCUSSIONS

A. Experimental Setup

In order to validate the performance of the nonlinear adaptive inverse controller for hysteresis compensation, an experimental platform is established as shown in Fig. 19. The experimental platform is set up on an optical table and the experiments were carried out in a Class 1000 clean room. The laser displacement sensor was employed to measure the output displacement of the proposed stage. The signal was acquired by a dynamic signal acquisition module (NI PXI-4472, made by National Instru-

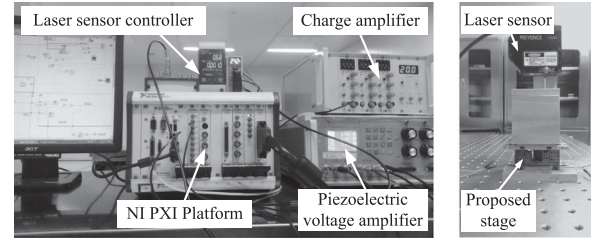


Fig. 19. Experimental platform.

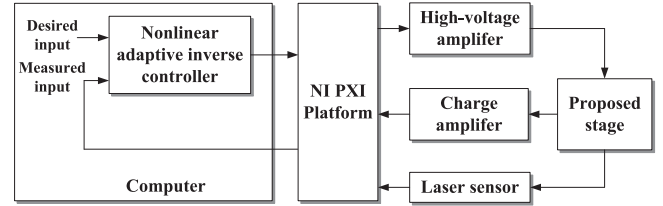


Fig. 20. Structure of the experimental system.

ments Corporation) with 24-bit resolution ADCs and 102.4 kS/s sampling rate. PXI Platform (NI PXI-8186, made by National Instruments Corporation) equipped with LabVIEW 8.6 system was used to achieve the real-time control. The control signal was outputted by a high-speed analog output module (NI PXI-6733, made by National Instruments Corporation) with 16-bit resolution and 10 MS/s update rate, and then amplified by a piezoelectric voltage amplifier (Rhvd-3C, made by RongZhi-NaXin Corporation) before applying to the proposed stage. Key features of this amplifier include -20 V to 200 V output voltage range, 20 V/V dc voltage gain, up to 50 -kHz signal bandwidth, and less than 2 -mV root-mean-square (RMS) noise. The sampling frequency in the control experiments is set to 1000 Hz. Both the input voltage and the feedback displacement are acquired and stored by a computer with Intel Core i3 2.93-GHz processor and 2-GB RAM. The whole structure of the experimental system is illustrated in Fig. 20.

B. Results and Discussion

To demonstrate the performance of the proposed nonlinear adaptive inverse controller for hysteresis compensation, the position tracking for periodic triangular input trajectories with peak-to-peak amplitude of $6\text{ }\mu\text{m}$ are experimentally tested. Such input trajectories have been used for micropositioning systems such as tip motion under the scanning mode of AFM [38]. Because the output voltage range of the piezoelectric voltage amplifier is from -20 V to 200 V, the smart piezo stack can not generate enough negative displacement driven by the limited negative voltage. In order to overcome this problem, 100 -V offset voltage was applied to the smart piezo stack. Owing to this offset voltage, the proposed stage is capable of generating positive displacement when the applied voltage is greater than 100 V, and generating negative displacement when the applied voltage is less than 100 V. Because analog output module with 16-bit resolution and piezoelectric voltage amplifier with 20 V/V dc voltage gain are employed, the theoretical displacement

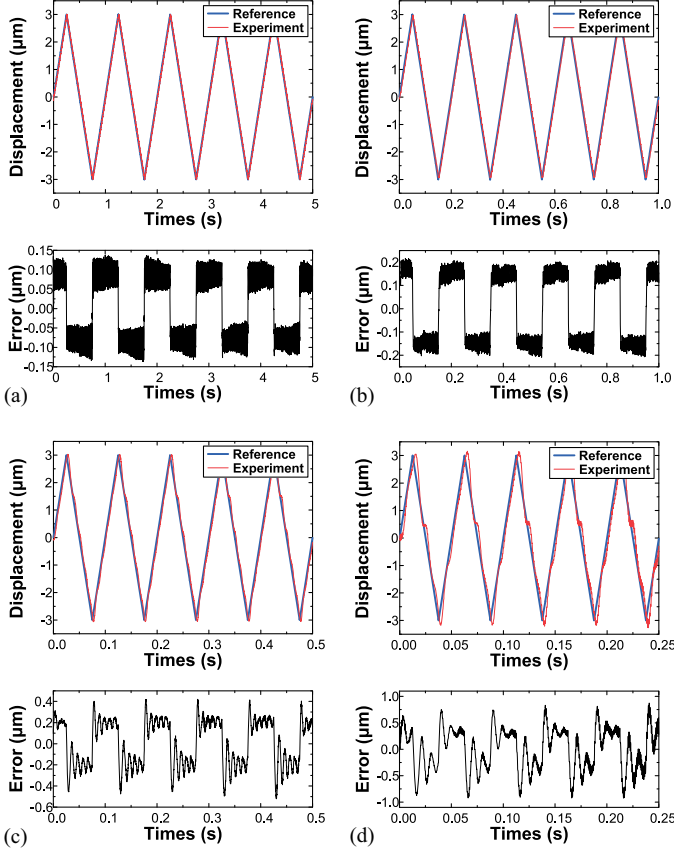


Fig. 21. Position tracking of triangular reference trajectories with amplitude of 6 μm and frequencies of: (a) 1 Hz, (b) 5 Hz, (c) 10 Hz, and (d) 20 Hz.

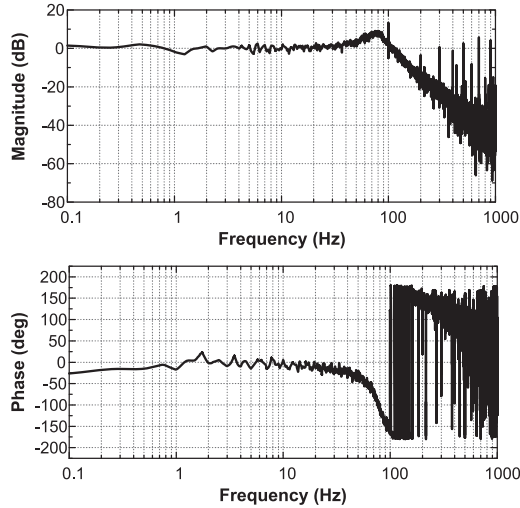


Fig. 22. Closed-loop frequency response measured from the applied reference to the resulting displacement.

resolution of the smart piezo stack is less than 0.5 nm. In Fig. 21, position tracking of triangular reference trajectories with frequencies of 1, 5, 10, and 20 Hz is shown. Moreover, the closed-loop frequency response measured from the applied reference to the resulting displacement is shown in Fig. 22. With the aid of the proposed intelligent control algorithm, the structure of the online adaptive inverse controller kept rapidly

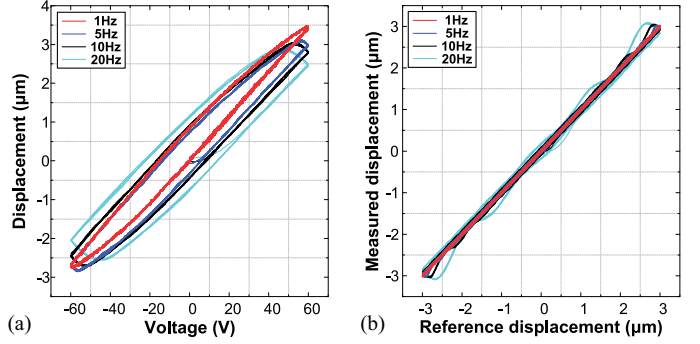


Fig. 23. Hysteresis plots of the proposed stage: (a) before compensation control, and (b) after compensation control.

TABLE IV
TRACKING PERFORMANCE OF TRIANGULAR REFERENCE TRAJECTORIES WITH VARIOUS FREQUENCIES

Frequency (Hz)	1	5	10	20
$e_{\text{rms}}(\%)$	1.43	2.54	3.72	6.72

changing during the entire control process. The experimental results demonstrate that in a self-sensing way without the aid of commercial displacement sensor, the proposed stage is capable of accurately tracking triangular reference signals with various frequencies regardless of the rate-dependent hysteresis phenomenon. In Fig. 23, the hysteresis plots of the proposed stage before and after compensation control are shown for comparison.

Define the position tracking RMS error e_{rms} as a percentage of the total output range as

$$e_{\text{rms}}(\%) = \left[\frac{\sqrt{\frac{1}{N_t} \sum_{k=1}^{N_t} (\bar{y}_d(k) - \bar{y}(k))^2}}{\max(\bar{y}_d(k)) - \min(\bar{y}_d(k))} \right] \times 100\% \quad (20)$$

where N_t is the number of test datasets in the experiment. The tracking performance of triangular reference trajectories with various frequencies are summarized in Table IV. It can be seen that the RMS error increases with the frequency of triangular reference signal. This is mainly because the nonlinear adaptive inverse controller utilizes the intelligent learning algorithm to compensate for the hysteresis in the proposed stage. The more information the intelligent controller can obtain from the dynamics of the plant, the more excellent performance it can reach. However, when the frequency of triangular reference signal increases, the information obtained during each period reduces because of the fixed sampling rate. Therefore, using hardware implementation with higher performance is expected to increase the system resolution for reference signal with higher frequency.

VI. CONCLUSION AND FUTURE WORK

This paper has presented the design, modeling, calibration, and hysteresis compensation of a monolithic self-sensing precision stage for active vibration isolation which features a compact structure without employing additional commercial sensors. The active part of the proposed stage is used for position control, vibration control, and dynamics monitoring. Based on the

functional requirements, a smart piezo stack which can be used as a displacement actuator, a force sensor, and a displacement sensor is designed and fabricated. In order to obtain the accurate relationship between the experienced force and the measured voltage, force sensing characteristics of the smart piezo stack is calibrated via experiments. In addition, we reveal and quantitatively analyze the crosstalk phenomenon in the smart piezo stack. According to the results of FEA and a group of experiments, it turns out that the crosstalk voltage account for less than 8% of the sensor voltage in the smart piezo stack. Furthermore, according to the model of the smart piezo stack and the analysis of the crosstalk phenomenon, we establish the dynamics model of the active part of the proposed stage, which demonstrate that the proposed stage is capable of achieving displacement measurement by itself with the aid of the embedded smart piezo stack. Aiming to effectively compensate for the rate-dependent hysteresis in the proposed stage, we develop a nonlinear adaptive inverse controller. Compared with most existing controllers for hysteresis compensation, a distinctive feature of the developed controller is that it can be designed even when the dynamics of proposed stage is unknown. Experimental results show that the active part of the proposed stage successfully tracks the triangular reference signal with various frequencies without dynamics modeling and additional commercial sensors.

In the future, we plan to utilize three proposed stages which have already been fabricated to construct a multidegrees of freedom (DOF)s precision active vibration isolation system. Taking advantage of the smart piezo stack which is capable of accomplishing force measurement, displacement measurement, and precision position control, we are quite interested in designing an intelligent control algorithm based on the support vector machine (SVM) [31] to extend the adaptive inverse control scheme to multi-DOFs platform for achieving both position control and adaptive vibration suppression. Compared with artificial neural network, the SVM exhibits the advantages of global optimization and higher generalization capability [39]. In addition, as the calculation efficiency and accuracy strongly depend on the hardware implement of the proposed intelligent control algorithm, we intend to adopt a superior real-time control system to further enhance the system performance.

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REFERENCES

- [1] P. C. Chen and M. C. Shih, "Robust control of a novel active pneumatic vibration isolator through floor vibration observer," *J. Vib. Control*, vol. 17, pp. 1325–1336, 2011.
- [2] M. Kim, H. Kim, and D. G. Gweon, "Design and optimization of voice coil actuator for six degree of freedom active vibration isolation system using halfbach magnet array," *Rev. Sci. Instrum.*, vol. 83, pp. 105117–1–105117-9, 2012.
- [3] J. Holterman and T. J. de Vries, "Active damping based on decoupled collocated control," *IEEE/ASME Trans. Mechatronics*, vol. 10, no. 2, pp. 135–145, Apr. 2005.
- [4] Y. Li, J. Huang, and H. Tang, "A compliant parallel XY micromotion stage with complete kinematic decoupling," *IEEE Trans. Autom. Sci. Eng.*, vol. 9, no. 3, pp. 538–553, Jul. 2012.
- [5] E. Hoque, T. Mizuno, D. Kishita, M. Takasaki, and Y. Ishino, "Development of an active vibration isolation system using linearized zero-power control with weight support springs," *J. Vib. Acoust.*, vol. 132, no. 4, pp. 041006-1–041006-9, 2010.
- [6] Y. K. Yong, A. J. Fleming, and S. R. Moheimani, "A novel piezoelectric strain sensor for simultaneous damping and tracking control of a high-speed nanopositioner," *IEEE/ASME Trans. Mechatronics*, vol. 18, no. 3, pp. 1113–1121, Jun. 2013.
- [7] D. O. Lee, L. H. Kang, and J. H. Han, "Active vibration isolation demonstration system using the piezoelectric unimorph with mechanically prestressed substrate," *J. Intell. Mater. Syst. Struct.*, vol. 22, no. 13, pp. 1399–1409, 2011.
- [8] A. J. Fleming, "Nanopositioning system with force feedback for high-performance tracking and vibration control," *IEEE/ASME Trans. Mechatronics*, vol. 15, no. 3, pp. 433–447, Jun. 2010.
- [9] L. Liu, K. Tan, S. Chen, C. Teo, and T. Lee, "Discrete composite control of piezoelectric actuators for high-speed and precision scanning," *IEEE Trans. Ind. Informat.*, vol. 9, no. 2, pp. 859–868, May 2013.
- [10] Q. Xu, Y. Li, and N. Xi, "Design, fabrication, and visual servo control of an XY parallel micromanipulator with piezo-actuation," *IEEE Trans. Autom. Sci. Eng.*, vol. 6, no. 4, pp. 710–719, Oct. 2009.
- [11] Y. Li and Q. Xu, "Design and robust repetitive control of a new parallel-kinematic XY piezostage for micro/nanomanipulation," *IEEE/ASME Trans. Mechatronics*, vol. 17, no. 6, pp. 1120–1132, Dec. 2012.
- [12] B. J. Kenton and K. K. Leang, "Design and control of a three-axis serial-kinematic high-bandwidth nanopositioner," *IEEE/ASME Trans. Mechatronics*, vol. 17, no. 2, pp. 356–369, Apr. 2012.
- [13] Y. K. Yong, S. S. Aphale, and S. R. Moheimani, "Design, identification, and control of a flexure-based XY stage for fast nanoscale positioning," *IEEE Trans. Nanotechnol.*, vol. 8, no. 1, pp. 46–54, Jan. 2009.
- [14] A. J. Fleming and K. K. Leang, "Integrated strain and force feedback for high-performance control of piezoelectric actuators," *Sens. Actuat. A, Phys.*, vol. 161, no. 1, pp. 256–265, 2010.
- [15] S. S. Aphale, A. J. Fleming, and S. R. Moheimani, "Integral resonant control of collocated smart structures," *Smart Mater. Struct.*, vol. 16, no. 2, pp. 439–446, 2007.
- [16] A. J. Fleming, "A review of nanometer resolution position sensors: Operation and performance," *Sens. Actuat. A, Phys.*, vol. 190, no. 1, pp. 106–126, 2013.
- [17] P. Ronkanen, P. Kallio, M. Vilkko, and H. N. Koivo, "Displacement control of piezoelectric actuators using current and voltage," *IEEE/ASME Trans. Mechatronics*, vol. 16, no. 1, pp. 160–166, Feb. 2011.
- [18] B. Bhikkaji, M. Ratnam, A. J. Fleming, and S. R. Moheimani, "High-performance control of piezoelectric tube scanners," *IEEE Trans. Control Syst. Technol.*, vol. 15, no. 5, pp. 853–866, Sep. 2007.
- [19] A. J. Fleming, S. S. Aphale, and S. R. Moheimani, "A new method for robust damping and tracking control of scanning probe microscope positioning stages," *IEEE Trans. Nanotechnol.*, vol. 9, no. 4, pp. 438–448, Jul. 2010.
- [20] B. Mokaberi and A. A. Requicha, "Compensation of scanner creep and hysteresis for AFM nanomanipulation," *IEEE Trans. Autom. Sci. Eng.*, vol. 5, no. 2, pp. 197–206, Apr. 2008.
- [21] K. K. Leang and S. Devasia, "Feedback-linearized inverse feedforward for creep, hysteresis, and vibration compensation in AFM piezoactuators," *IEEE Trans. Control Syst. Technol.*, vol. 15, no. 5, pp. 927–935, Sep. 2007.
- [22] Y. Li and Q. Xu, "A totally decoupled piezo-driven XYZ flexure parallel micropositioning stage for micro/nanomanipulation," *IEEE Trans. Autom. Sci. Eng.*, vol. 8, no. 2, pp. 265–279, Apr. 2011.
- [23] Y. K. Yong and T. F. Lu, "Kinestatic modeling of 3-RRR compliant micro-motion stages with flexure hinges," *Mech. Mach. Theory*, vol. 44, no. 6, pp. 1156–1175, 2009.
- [24] Q. Xu, "New flexure parallel-kinematic micropositioning system with large workspace," *IEEE Trans. Robot.*, vol. 28, no. 2, pp. 478–491, Apr. 2012.
- [25] Q. Xu, "Design and development of a flexure-based dual-stage nanopositioning system with minimum interference behavior," *IEEE Trans. Autom. Sci. Eng.*, vol. 9, no. 3, pp. 554–563, Jul. 2012.
- [26] Q. Xu, "Design and development of a compact flexure-based xy precision positioning system with centimeter range," *IEEE Trans. Ind. Electron.*, vol. 61, no. 2, pp. 893–903, Feb. 2014.

- [27] D. W. Schubert, A. M. Beard, S. F. Shedd, M. R. Earles, Jr., and A. H. Von Flotow, "Stiff actuator active vibration isolation system," U.S. Patent 5 660 255, Aug. 26, 1997.
- [28] H. Adriaens, W. L. De Koning, and R. Banning, "Modeling piezoelectric actuators," *IEEE/ASME Trans. Mechatronics*, vol. 5, no. 4, pp. 331–341, Dec. 2000.
- [29] G. Gu, L. Zhu, C. Su, and H. Ding, "Motion control of piezoelectric positioning stages: Modeling, controller design, and experimental evaluation," *IEEE/ASME Trans. Mechatronics*, vol. 18, no. 5, pp. 1459–1471, Oct. 2013.
- [30] G. Y. Gu and L. M. Zhu, "Modeling of rate-dependent hysteresis in piezoelectric actuators using a family of ellipses," *Sens. Actuat. A, Phys.*, vol. 165, no. 2, pp. 303–309, 2010.
- [31] P. K. Wong, Q. Xu, C. M. Vong, and H. C. Wong, "Rate-dependent hysteresis modeling and control of a piezostage using online support vector machine and relevance vector machine," *IEEE Trans. Ind. Electron.*, vol. 59, no. 4, pp. 1988–2001, Apr. 2012.
- [32] G. Y. Gu, M. J. Yang, and L. M. Zhu, "Real-time inverse hysteresis compensation of piezoelectric actuators with a modified Prandtl-Ishlinskii model," *Rev. Sci. Instrum.*, vol. 83, no. 6, pp. 065106-1–065106-8, 2012.
- [33] Y. Gao and M. J. Er, "NARMAX time series model prediction: Feedforward and recurrent fuzzy neural network approaches," *Fuzzy Sets Syst.*, vol. 150, no. 2, pp. 331–350, 2005.
- [34] S. Chen and S. A. Billings, "Representations of non-linear systems: The NARMAX model," *Int. J. Control*, vol. 49, no. 3, pp. 1013–1032, 1989.
- [35] Q. Li, J. Y. Yu, B. C. Mu, and X. D. Sun, "BP neural network prediction of the mechanical properties of porous NiTi shape memory alloy prepared by thermal explosion reaction," *Mater. Sci. Eng., A*, vol. 419, no. 1, pp. 214–217, 2006.
- [36] G. L. Plett, "Adaptive inverse control of linear and nonlinear systems using dynamic neural networks," *IEEE Trans. Neural Netw.*, vol. 14, no. 2, pp. 360–376, Mar. 2003.
- [37] B. Widrow and E. Walach, "Adaptive inverse control," *Control Eng. Pract.*, vol. 5, no. 1, pp. 146–147, 1997.
- [38] J. Yi, S. Chang, and Y. Shen, "Disturbance-observer-based hysteresis compensation for piezoelectric actuators," *IEEE/ASME Trans. Mechatronics*, vol. 14, no. 4, pp. 456–464, Aug. 2009.
- [39] Q. Xu, "Identification and compensation of piezoelectric hysteresis without modeling hysteresis inverse," *IEEE Trans. Ind. Electron.*, vol. 60, no. 9, pp. 3927–3937, Sep. 2013.



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